

L9



L9: Arithmetic Functions and Multiplicativity

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- A comment about notation: $\sum_{d|n, d>0} f(d)$ means the

summation of f over all distinct positive divisors d of n . For example,

$$\sum_{\substack{d|12 \\ d>0}} f(d) = f(1) + f(2) + f(3) + f(4) + f(6) + f(12).$$

Theorem 3.1: Let f be an arithmetic function, and for $n \in \mathbb{Z}$ with $n > 0$, let

$$F(n) = \sum_{d|n, d>0} f(d)$$

If f is multiplicative, then F is multiplicative.

Proof: Let m and n be relatively prime positive integers. To prove that F is multiplicative, we must prove that $F(mn) = F(m)F(n)$. We have that

$$F(mn) = \sum_{d|mn, d>0} f(d). \quad \text{Claim: Since } (m, n) = 1,$$

(2)

each divisor $d > 0$ of mn can be written uniquely as $d_1 d_2$ where $d_1, d_2 > 0$, $d_1 | m$, $d_2 | n$, and $(d_1, d_2) = 1$, and each such product $d_1 d_2$ corresponds to a divisor of mn . The case when either $m = 1$ or $n = 1$ are trivial. Assume that $m > 1$ and $n > 1$,

$$\text{and let } m = p_1^{a_1} p_2^{a_2} \dots p_r^{a_r}$$

$$n = q_1^{b_1} q_2^{b_2} \dots q_s^{b_s}$$

be the prime factorisations of m and n respectively. Since $(m, n) = 1$, all p_i and q_j are distinct. Let d be a positive divisor of mn . Then

$$d = p_1^{e_1} p_2^{e_2} \dots p_r^{e_r} q_1^{f_1} q_2^{f_2} \dots q_s^{f_s},$$

where the integers e_i and f_j satisfy

$$0 \leq e_i \leq a_i \text{ and } 0 \leq f_j \leq b_j \text{ for all } i$$

$$\text{and } j. \text{ Now } d_1 = p_1^{e_1} p_2^{e_2} \dots p_r^{e_r} \text{ and}$$

$$d_2 = q_1^{f_1} q_2^{f_2} \dots q_s^{f_s} \text{ are as desired. The fact}$$

that each such product corresponds to a 3
divisor of d is clear. This proves the claim.
Then,

$$\begin{aligned} F(mn) &= \sum_{\substack{d|mn \\ d>0}} f(d) \\ &= \sum_{\substack{d_1|m, d_2|n \\ d_1, d_2>0}} f(d_1 d_2) \\ &= \sum_{\substack{d_1|m, d_2|n \\ d_1, d_2>0}} f(d_1) f(d_2) \quad (\text{since } f \text{ is multiplicative}) \\ &= \left(\sum_{\substack{d_1|m \\ d_1>0}} f(d_1) \right) \cdot \left(\sum_{\substack{d_2|n \\ d_2>0}} f(d_2) \right) \\ &= F(m) F(n), \end{aligned}$$

as desired. □

Example: Let $m=3$ and $n=4$, and keep the notation the same as Theorem 3.1. We show

(4)

that $F(3 \cdot 4) = F(3)F(4)$. We have

$$\begin{aligned}
 F(3 \cdot 4) &= \sum_{d|12, d>0} f(d) \\
 &= f(1) + f(2) + f(3) + f(4) + f(6) + f(12) \\
 &= f(1) + f(2) + f(4) + f(3) + f(6) + f(12) \\
 &= f(1 \cdot 1) + f(1 \cdot 2) + f(1 \cdot 4) + f(3 \cdot 1) \\
 &\quad + f(3 \cdot 2) + f(3 \cdot 4) \\
 &= f(1)f(1) + f(1)f(2) + f(1)f(4) \\
 &\quad + f(3)f(1) + f(3)f(2) + f(3)f(4) \\
 &= (f(1) + f(3))(f(1) + f(2) + f(4)) \\
 &= \left(\sum_{d_1|3, d_1>0} f(d_1) \right) \cdot \left(\sum_{d_2|4, d_2>0} f(d_2) \right) \\
 &= F(3)F(4),
 \end{aligned}$$

as desired.

The Euler ϕ -function

Theorem 3.2 : The Euler ϕ -function is multiplicative.

Proof: Let m and n be relatively prime (5)
 positive integers. We must prove that
 $\phi(mn) = \phi(m)\phi(n)$. We display the positive integers
 not exceeding mn as a matrix:

$$\begin{array}{cccccc}
 1 & m+1 & 2m+1 & \cdots & (n-1)m+1 \\
 2 & m+2 & 2m+2 & \cdots & (n-1)m+2 \\
 \vdots & \vdots & \vdots & \ddots & \vdots \\
 i & m+i & 2m+i & \cdots & (n-1)m+i \\
 \vdots & \vdots & \vdots & \ddots & \vdots \\
 m & 2m & 3m & \cdots & nm
 \end{array}$$

Consider the i th row of the matrix above. If
 $(m, i) > 1$, then no entry of the i th row is
 relatively prime to m , and so not relatively
 prime to mn (every entry of the i th row is
 of the form $km+i$ where $k \in \mathbb{Z}$, so if
 $(m, i) = d > 1$, then $d|m$ and $d|i$ implies that
 $d|km+i$ by Proposition 1.2). Hence, to determine
 how many integers in the matrix that are

relatively prime to mn , which is $\phi(mn)$, it suffices to examine the i th row of the matrix if and only if $(m, i) = 1$. There are $\phi(m)$ such rows. Now, given such an i th row, we must determine how many integers are relatively prime to mn . The entries of such a row are

$$i, m+i, 2m+i, \dots, (n-1)m+i.$$

Since $(m, i) = 1$, each of these integers are relatively prime to m . Furthermore, these n integers form a complete residue system modulo n .

Otherwise, $jm+i \equiv km+i \pmod{n}$ for some $0 \leq j, k \leq n-1$ with $j \neq k$. Then $jm \equiv km \pmod{n}$, and since $(m, n) = 1$, the inverse of m modulo n exists, say m' , and we have $jmm' \equiv kmm' \pmod{n}$, and so $j \equiv k \pmod{n}$. Impossible! Thus exactly $\phi(n)$ of these integers are relatively prime to n . Since these $\phi(n)$ integers are relatively prime to m , they are relatively prime to mn . In summary, exactly $\phi(m)$ rows of the matrix

Contain integers relatively prime to m . Each (7)
 Such row contains exactly $\phi(n)$ integers relatively
 prime to mn . Thus $\phi(mn) = \phi(m)\phi(n)$, as
 required. $\square$

• Since ϕ is multiplicative, it is determined by
 its values at powers of prime numbers.

Theorem 3.3: Let p be a prime number and let
 $a \in \mathbb{Z}$ with $a \geq 0$. Then

$$\phi(p^a) = p^a - p^{a-1}.$$

Proof: The total number of positive integers not
 exceeding p^a is p^a . The positive integers not
 exceeding p^a that are not relatively prime to p^a
 are precisely the multiples of p :

$$p, 2p, 3p, \dots, (p^{a-1})p.$$

There are p^{a-1} such multiples, and so we conclude
 that $\phi(p^a) = p^a - p^{a-1}$. $\square$

Theorem 3.4: Let $n \in \mathbb{Z}$ with $n > 0$. Then

$$\phi(n) = n \prod_{p \mid n, p \text{ prime}} \left(1 - \frac{1}{p}\right).$$

• The notation $\prod_{p|n, p \text{ prime}}$ means the product is (8)

taken over all distinct prime divisors p of n .

For example,

$$\prod_{p|12, p \text{ prime}} \left(1 - \frac{1}{p}\right) = \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right).$$

Proof (of Theorem 3.4): The result is clear for $n = 1$ (with the convention that the empty product is taken to be 1). Assume that $n > 1$ and that $n = p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r}$ with $p_1, p_2, \dots, p_r$ distinct prime numbers and $a_1, \dots, a_r$ positive integers. Then

$$\begin{aligned}\phi(n) &= \phi(p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r}) \quad \text{by Theorem 3.2} \\ &= \phi(p_1^{a_1}) \phi(p_2^{a_2}) \cdots \phi(p_r^{a_r}) \quad \text{by Theorem 3.3} \\ &= (p_1^{a_1} - p_1^{a_1-1})(p_2^{a_2} - p_2^{a_2-1}) \cdots (p_r^{a_r} - p_r^{a_r-1}) \\ &= p_1^{a_1} \left(1 - \frac{1}{p_1}\right) p_2^{a_2} \left(1 - \frac{1}{p_2}\right) \cdots p_r^{a_r} \left(1 - \frac{1}{p_r}\right) \\ &= p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r} \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_r}\right) \\ &= n \prod_{p|n} \left(1 - \frac{1}{p}\right).\end{aligned}$$

Example: Compute $\phi(504)$. We have $504 = 2^3 3^2 7$, (9)

$$\begin{aligned}\text{so } \phi(504) &= 504 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{7}\right) \\ &= 504 \left(\frac{1}{2}\right) \left(\frac{2}{3}\right) \left(\frac{6}{7}\right) \\ &= 144.\end{aligned}$$

Theorem 3.5 (Gauss): Let $n \in \mathbb{Z}$ with $n > 0$. Then

$$\sum_{d|n, d>0} \phi(d) = n.$$

Proof: Let d be a positive divisor of n and put

$$S_d := \{m \in \mathbb{Z} : 1 \leq m \leq n, (m, n) = d\}.$$

Now $(m, n) = d$ if and only if $\left(\frac{m}{d}, \frac{n}{d}\right) = 1$.

Thus the number of integers in S_d is equal to the number of positive integers not exceeding n/d that are relatively prime to n/d i.e.

$|S_d| = \phi(n/d)$. Since every integer from 1 to n inclusive belongs to exactly one S_d , we have

that $n = \sum_{d|n, d>0} \phi(n/d)$. As d runs through

all positive divisors of n , so to does n/d , (10)
thus

$$n = \sum_{d|n, d>0} \phi(n/d) = \sum_{d|n, d>0} \phi(d),$$

as desired. $\square$

Example: Let $n=12$ and all the notation be as in Theorem 3.5. We must show that

$$\sum_{d|12, d>0} \phi(d) = 12.$$

We have

$$S_1 = \{1, 5, 7, 11\}, \quad |S_1| = 4 = \phi(12)$$

$$S_2 = \{2, 10\}, \quad |S_2| = 2 = \phi(6)$$

$$S_3 = \{3, 9\}, \quad |S_3| = 2 = \phi(4)$$

$$S_4 = \{4, 8\}, \quad |S_4| = 2 = \phi(3)$$

$$S_6 = \{6\}, \quad |S_6| = 1 = \phi(2)$$

$$S_{12} = \{12\}, \quad |S_{12}| = 1 = \phi(1)$$

In all cases we have that $|S_d| = \phi(12/d)$.
Since each integer from 1 to 12 inclusive

belongs to exactly one S_d we have that (11)

$$12 = \phi(1) + \phi(2) + \phi(3) + \phi(4) + \phi(6) + \phi(12).$$